



Telco Customer Churn Analysis



Project Overview

Customer churn is an important business problem for telecommunications companies because losing existing customers can affect recurring revenue and long-term customer relationships.

This project performs an **Exploratory Data Analysis (EDA)** on a Telco Customer Churn dataset to understand customer characteristics, service subscriptions, contract information, payment methods, tenure, and their relationship with customer churn.

The analysis was performed using **Python and popular data analytics libraries**, with a strong focus on data cleaning, categorical analysis, statistical exploration, and visualization.



Project Objectives

The main objectives of this project are:

- Understand the structure and characteristics of the customer dataset.
 - Perform data quality checks and prepare the data for analysis.
 - Analyze the overall distribution of customer churn.
 - Understand customer behavior across different demographic groups.
 - Examine how customer tenure is distributed.
 - Analyze churn across different contract types.
 - Study the relationship between internet and additional services and churn.
 - Analyze payment methods and their relationship with customer churn.
 - Compare customer segments using percentage-based visualizations.
 - Generate business-oriented insights from the exploratory analysis.
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Dataset

The dataset contains customer-level information related to:



Customer Information

- Customer ID
- Gender

- Senior Citizen
- Partner
- Dependents



Customer Relationship

- Tenure
- Contract
- Payment Method



Services

- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies



Billing Information

- Monthly Charges
- Total Charges



Target Variable

- Churn

The dataset contains **7,043 customer records and 21 features**.

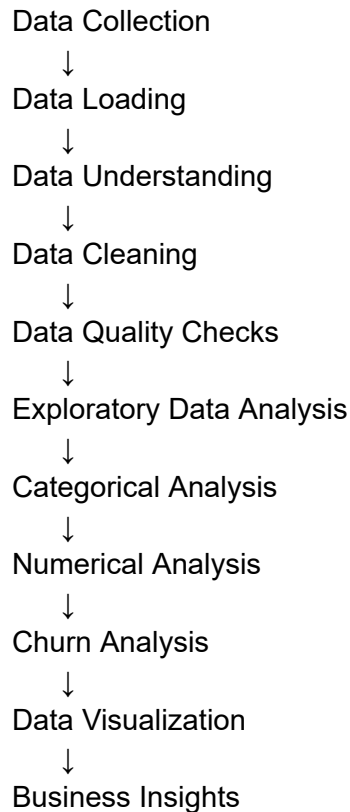


Technologies Used

Technology	Purpose
Python	Data analysis and programming
Pandas	Data manipulation and analysis
NumPy	Numerical operations

Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter Notebook	Analysis environment

Project Workflow



Data Cleaning & Preparation

The initial stage of the project focused on understanding and preparing the dataset.

The following activities were performed:

- Loaded the dataset using Pandas.
- Inspected the number of rows and columns.
- Examined column names and data types.
- Checked for missing values.
- Checked for duplicate records.

- Reviewed categorical variables.
- Examined numerical variables.
- Prepared variables for visualization and analysis.
- Converted/handled relevant data types where required.

Data cleaning was performed before visualization to improve the reliability of the analysis.



Exploratory Data Analysis

1. Customer Churn Distribution

The first step was to understand the distribution of customers who:

- Stayed with the company
- Churned from the company

A count plot was used to visualize the overall churn distribution.

Purpose

This provides a high-level understanding of the target variable and helps identify whether churn is evenly distributed or whether one class contains substantially more customers.

2. Gender Analysis

Customer churn was analyzed across gender categories.

Analysis Question

Does customer churn distribution differ between male and female customers?

A count-based visualization was used to compare the groups.

This analysis helps understand whether gender is associated with noticeably different customer behavior in the dataset.

3. Senior Citizen Analysis

The `SeniorCitizen` variable was analyzed to compare churn patterns between:

- Senior citizens
- Non-senior citizens

A count plot was used to visualize the customer distribution.

Business Relevance

Customer demographics can help businesses understand whether particular customer groups require different retention strategies.



4. Tenure Analysis

`tenure` represents the length of time a customer has stayed with the company.

The analysis examined the distribution of customer tenure and its relationship with churn.

Key Questions

- How long do customers typically stay?
- Are newer customers represented differently from long-term customers?
- Does churn behavior vary across tenure levels?

Histograms/count-based visualizations were used to explore tenure.



5. Contract Analysis

Contract type was analyzed against customer churn.

The dataset contains contract categories such as:

- Month-to-month
- One year
- Two year

A categorical visualization was used to compare churn across contract categories.

Business Relevance

Contract structure is an important customer-retention factor because customers with different commitment periods may exhibit different churn behavior.



6. Internet Service Analysis

The **InternetService** variable was analyzed to understand customer distribution across internet-service categories and their churn behavior.

The analysis helps answer:

How does churn vary across different internet service types?



7. Additional Service Analysis

The following service-related variables were analyzed individually:

PhoneService
MultipleLines
InternetService
OnlineSecurity
OnlineBackup
DeviceProtection
TechSupport
StreamingTV
StreamingMovies

Count plots were created to understand customer adoption across these services.

Why this matters

Additional services can provide useful information about customer engagement with the company's products.



8. Payment Method Analysis

Customer payment methods were analyzed in relation to churn.

The analysis examines whether customers using different payment methods show different churn distributions.

Business Question

Is there a noticeable difference in churn behavior across payment methods?

This can help identify customer segments that may require additional investigation.



9. Percentage-Based Churn Analysis

In addition to simple count plots, percentage-based analysis was used to make comparisons between categories easier.

A **100% stacked bar chart** can be used to show the proportion of:

Stayed

vs.

Churned

within each category.

For example:

Contract Type



Month-to-month → Stayed % | Churned %

One year → Stayed % | Churned %

Two year → Stayed % | Churned %

Why Percentage Analysis?

Raw counts can sometimes be misleading when category sizes are different.

Percentage-based visualization allows the churn proportion within each category to be compared more directly.



Visualizations Used

The project includes several visualization techniques:

- Count plots
- Bar charts
- Histograms
- Pie charts
- Stacked bar charts
- Percentage-based stacked bar charts

These visualizations were selected according to the type of variable and analytical question.



Business Insights Framework

The analysis focuses on identifying relationships between churn and:

Customer Profile

- Gender
- Senior Citizen
- Partner
- Dependents

Customer Relationship

- Tenure
- Contract

Services

- Internet Service
- Online Security
- Online Backup
- Device Protection

- Tech Support
- Streaming Services

Payment

- Payment Method

These dimensions can be used by a telecom business to identify customer segments that deserve further investigation for retention.

Note: EDA shows associations and patterns in the dataset. It does not by itself establish that a particular factor causes churn.



Skills Demonstrated

This project demonstrates practical experience in:

Python

- Data loading
- Data manipulation
- Conditional analysis
- Basic statistical exploration

Pandas

- DataFrame inspection
- Filtering
- Grouping
- Aggregation
- Data cleaning
- Categorical analysis

NumPy

- Numerical calculations
- Percentage calculations
- Data transformation

Matplotlib & Seaborn

- Statistical visualization
- Categorical plots
- Distribution plots
- Comparative visualization
- Subplots
- Stacked bar charts

Data Analytics

- Exploratory Data Analysis
 - Data quality checking
 - Customer segmentation
 - Churn analysis
 - Business-oriented interpretation
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Project Structure

Telco-Customer-Churn-Analysis/

```
|  
├── Telco_Customer_Churn_Analysis.ipynb  
├── WA_Fn-UseC_-Telco-Customer-Churn.csv  
└── README.md
```



Future Improvements

The current project focuses mainly on exploratory analysis. It can be extended by:

- Creating a Power BI churn dashboard.
 - Building a churn prediction model.
 - Performing feature engineering.
 - Encoding categorical variables.
 - Comparing machine-learning classification models.
 - Evaluating models using precision, recall, F1-score, and ROC-AUC.
 - Performing deeper customer segmentation.
 - Developing an interactive business dashboard.
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Project Outcome

This project provided practical experience in converting a raw customer dataset into meaningful exploratory insights.

The complete workflow demonstrates how a data analyst can move from:

Raw Data → Cleaning → Exploration → Visualization → Pattern Identification → Business Understanding

It also demonstrates hands-on experience with **Python, Pandas, NumPy, Matplotlib, Seaborn, and Exploratory Data Analysis**.



Author

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Skills: Python | SQL | Excel | Power BI | Tableau | Pandas | NumPy | Data Visualization



If you find this project useful

Feel free to explore the notebook, review the analysis, and use the project as a reference for learning **Exploratory Data Analysis with Python**.